

NIKILA KARPURAM

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Professional Summary

- Certified Cloud Data Engineer with 6 years of experience in building scalable, secure, and high-performance data solutions.
- Proficient in cloud services across AWS, GCP, and Azure, with deep expertise in data pipeline development and ETL orchestration.
- Strong programming skills in Python and SQL, with a background in data wrangling, transformation, and machine learning.
- Experienced in optimizing the performance of Redshift, BigQuery, and Snowflake warehouses, achieving significant query speedups.
- Adept at implementing real-time data ingestion and analytics using Kafka, Kinesis, Pub/Sub, and Spark Streaming.
- Built AI-powered chatbots and implemented NLP solutions for financial insights and automation.
- Led modernization of legacy systems to cloud-native architectures, reducing infrastructure costs by up to 50%.
- Delivered interactive business intelligence dashboards using Power BI, Tableau, and Looker to drive data-informed decisions.
- Implemented automated data quality and anomaly detection systems to ensure reliable, accurate reporting.
- Strong collaborative skills, having worked with cross-functional teams across data science, engineering, and business units.

Technical Skills

- Cloud Platforms: AWS (Redshift, Lambda, S3, Glue, EC2, ELB, VPC, RDS, AMI, IAM, EMR, Kinesis, Glue, Cloud Watch, Terraform, Amazon Aura, SNS, SQS, EBS, Route53, AWS Management Console, AWS CLI), GCP (BigQuery, Dataflow, Pub/Sub), Azure (Data Factory, Microsoft Azure Databricks, Data Lake, Blob Storage, Synapse, ML)
- Databases: SQL Database, SQL Data Warehouse, Cosmos DB, Azure Active Directory Databases MySQL, MS SQL SERVER, Snowflake NoSQL Database Cassandra, MongoDB, PostgreSQL, Oracle RDBMS
- Big Data Tools: Apache Spark, Kafka, Airflow, EMR, Snowflake, HDFS, Yarn, MapReduce, Hive, Airflow Hadoop Distributions Apache Hadoop 2.x/1.x
- Data Visualization: Power BI, Tableau, Looker

- DevOps & Tools: Docker, Kubernetes, Jenkins, Git, CI/CD, Terraform
- Other: Data Modeling, ETL, Data Quality, Data Governance, Machine Learning
- Operating systems Linux, Windows, Mac OS Software /Tools Microsoft Excel, Shell Scripting, Linux, Jupyter Notebook, PyCharm, Visual Studio Reporting Tools/ETL Tools Informatica, PySpark, HiveQL, Apache Spark

Professional Experience

Data Engineer | BlueScape LLC, Atlanta, GA

May 2024 – Present

- Designed and implemented ETL pipelines using Azure Data Factory and Databricks to manage large-scale healthcare data, optimizing system performance.
- Built and deployed scalable ETL/ELT pipelines using Python and JavaScript, automating data ingestion from APIs and third-party business platforms.
- Built and maintained Snowflake data models tailored to healthcare analytics, improving data availability and processing efficiency.
- Designed and optimized SQL queries in Oracle RDBMS to generate standardized and ad-hoc clinical study reports.
- Developed interactive Power BI dashboards and visualizations to support clinical trial metrics, performance indicators, and compliance reporting.
- Maintained and enhanced a robust reporting portfolio, improving query efficiency by 25% through performance tuning and restructuring.
- Conducted user training and created documentation to support business users in navigating and interpreting reports and dashboards.
- Contributed to the standardization of reporting conventions, improving data consistency and stakeholder trust in reporting outputs.
- Utilized strong analytical and problem-solving skills to identify data quality issues and recommend remediation steps in collaboration with data management teams.
- Demonstrated ability to manage multiple reporting projects simultaneously while maintaining compliance with quality systems and regulatory standards.
- Enhanced data workflows by utilizing Snowflake features such as clustering and partitioning, improving data retrieval times.
- Developed distributed data systems for real-time healthcare analytics with Azure Databricks and Delta Lake.
- Designed ingestion pipelines from DB2 and Teradata into Azure SQL using Data Factory, ensuring high-quality data transformation.
- Automated PySpark notebooks in Databricks for data extraction, cleansing, and ingestion, reducing manual intervention.
- Integrated Delta Lake with Azure Data Lake Storage for reliable, structured data storage supporting healthcare analytics.

- Improved real-time data access for patient monitoring systems using Databricks and Snowpipe.
- Managed non-relational databases like MongoDB and Cassandra for storing healthcare records and unstructured data.
- Created Power BI dashboards to visualize patient metrics, outcomes, and system usage, supporting data-driven healthcare delivery.
- Ensured HIPAA compliance by implementing robust data masking, logging, and audit strategies, safeguarding sensitive health

Data Engineer | Wipro Pvt Limited, India

Dec 2021 – Aug 2023

- Designed and maintained scalable ETL pipelines using AWS Glue, Redshift, and EMR, improving data processing throughput by 40% for enterprise analytics.
- Developed and deployed JavaScript-based scripts to automate data transformation and processing within web applications and serverless environments.
- Built and maintained API integrations to ingest data from SaaS platforms (e.g., CRM, marketing automation tools, financial systems) into cloud-based data warehouses.
- Engineered real-time data pipelines by leveraging RESTful APIs to synchronize and push data between systems, ensuring consistency and low latency.
- Reduced Redshift query latency from 10 seconds to under 3 seconds through distribution key tuning, sort key optimization, and workload management enhancements.
- Automated infrastructure provisioning using AWS CloudFormation, decreasing setup time and ensuring environment consistency across development and production.
- Built distributed data processing jobs with Apache Spark on EMR, enhancing batch pipeline performance for structured and semi-structured data.
- Integrated ML models into production using AWS SageMaker, improving supply chain demand forecasting accuracy.
- Created business intelligence dashboards in Tableau and Looker to support decision-making across marketing, finance, and operations teams.
- Leveraged AWS Step Functions and Lambda for orchestration of serverless workflows, reducing operational costs and improving maintainability.
- Developed robust CI/CD pipelines using AWS CodePipeline and PyTest, streamlining code deployment and increasing release velocity.
- Created parameterized queries and data models from Oracle RDBMS for integration into Power BI and SSRS dashboards.
- Conducted data validation and quality checks on Oracle-hosted datasets to support regulatory and audit compliance requirements.
- Maintained documentation for Oracle-based reporting solutions and contributed to the standardization of SQL coding practices across teams.

- Containerized data processing services using Docker and deployed via Kubernetes, improving scalability and reducing deployment errors.
- Implemented data security and compliance practices by enforcing IAM roles, encryption policies via KMS, and VPC-level access controls.

Data Engineer | Google, India

Dec 2020 – Sep 2021

- Designed and deployed end-to-end ETL workflows using Google Cloud Dataflow, Pub/Sub, and BigQuery to power real-time insurance analytics and reporting systems.
- Automated complex data transformation pipelines using Python and Apache Airflow, reducing manual intervention and increasing operational efficiency by 50%.
- Designed and maintained complex SQL queries and PL/SQL procedures within Oracle RDBMS to support data extraction, transformation, and reporting needs.
- Developed and optimized views, indexes, and stored procedures in Oracle to ensure high performance and data consistency across analytics solutions.
- Supported clinical trial data reporting by querying large datasets from Oracle databases, ensuring accuracy, integrity, and timeliness of operational insights.
- Collaborated with business analysts and clinical operations teams to translate requirements into efficient Oracle SQL logic and reusable data models.
- Managed schema updates, performance tuning, and troubleshooting within Oracle RDBMS environments, ensuring scalable and reliable data architecture.
- Developed real-time claim ingestion pipelines to support dynamic underwriting, fraud detection, and policy evaluation using streaming data from Pub/Sub.
- Integrated data across Snowflake and Redshift environments for unified analytics, enabling seamless cross-cloud insights and faster decision-making.
- Built scalable ETL solutions using Talend and Matillion to orchestrate batch and streaming jobs, ensuring reliability across diverse insurance data sources.
- Developed interactive dashboards with Power BI and Looker to visualize key insurance metrics such as loss ratios, claims frequency, and underwriting performance.
- Boosted BigQuery performance by 30% through query optimization techniques, including clustering, partitioning, and storage tuning.
- Partnered with underwriters and finance teams to build actionable, self-service data models that supported executive and regulatory reporting.
- Created anomaly detection pipelines using Python and statistical models to flag outliers and improve the quality and reliability of claims data.

Data Engineer | ICICI Bank, India

May 2019 – Dec 2020

- Designed and implemented high-performance ETL pipelines using Apache Spark and AWS Lambda, increasing data processing speed by 3x for large-scale financial datasets.
- Implemented secure API authentication mechanisms (OAuth 2.0, token-based auth) and standardized error handling for robust data ingestion.
- Collaborated with frontend and backend developers to integrate JS-based data services into existing applications, enhancing data-driven UI/UX features.
- Designed reusable JS modules for data formatting, validation, and enrichment before loading into data lakes or warehouses.
- Contributed to the optimization of ETL workflows involving API-based data sources, reducing data latency by 30% and improving operational efficiency.
- Migrated legacy on-premises systems to AWS Redshift and RDS, cutting infrastructure costs while improving scalability and accessibility for analytics teams.
- Developed interactive Power BI dashboards using DAX to create calculated columns, measures, and KPIs, enabling real-time tracking of business performance metrics.
- Wrote and optimized complex SQL queries across relational databases (e.g., Oracle, PostgreSQL, Snowflake) for data extraction, transformation, and reporting.
- Automated data processing and analysis workflows using Python (pandas, NumPy, matplotlib) to support data-driven decision-making in operational and clinical domains.
- Conducted statistical analysis and predictive modeling using R (tidyverse, caret, ggplot2), delivering insights for outcome-based strategy improvements.
- Integrated data from multiple sources using ETL pipelines, blending structured and semi-structured data for analytics-ready datasets.
- Applied DAX to implement dynamic filtering, time intelligence, and row-level security in Power BI reports for department-specific analytics.
- Collaborated with data scientists and analysts to develop reproducible Python/R notebooks for exploratory data analysis, visualizations, and hypothesis testing.
- Delivered insights to stakeholders through clean visual stories using a combination of SQL, R/Python analysis, and Power BI dashboards.
- Developed real-time fraud detection systems using AWS Kinesis and Apache Kafka, reducing fraudulent transactions through anomaly detection and alerting.
- Built a centralized financial data lake architecture with Amazon S3, Glue, and Athena, enhancing data availability and reducing department redundancy.
- Automated CI/CD processes using Terraform and AWS CodePipeline, decreasing deployment errors by 80% and accelerating release cycles for data pipelines.
- Created predictive models for credit scoring using Spark MLlib and Python, increasing loan approval accuracy and reducing default risks.
- Tuned Redshift clusters by optimizing sort and distribution keys, significantly improving query performance and reducing data storage costs.
- Automated compliance reporting aligned with GDPR and PCI-DSS, reducing manual efforts by 90% and ensuring audit readiness.

- Implemented AWS Glue Data Catalog to establish a centralized metadata repository, improving data discoverability and lineage across financial systems.

Projects

- AI-Powered Financial Chatbot (BCG GenAI Simulation, Feb 2025): Built chatbot in Python using NLP for parsing 10-K/10-Q filings.
- Accenture Data Analytics Simulation (Oct 2024): Cleaned and analyzed datasets; delivered business insights to social media client.
- Secure Cloud Auditing (Dec 2020 – Jun 2021): Developed a Java-based auditing scheme for secure, shared data integrity verification.

Education

M.S. in Information Technology Management, Computer Science – Concordia University, Saint Paul, MN

B.Tech in Computer Science – JNTUH, India

Certifications

AWS Certified Data Engineer